

Using MonetDB for High-Performance Applications

MonetDB is a new, column-store database paradigm for faster access to and high performance of applications.



In this era of Big Data, it is difficult to manage, store and manipulate data. To resolve this issue, there are different paradigms that organisations use to store data, such as structured, unstructured, semi-structured, row based and column based databases. In this article, we are going to explore a popular, open source, free column based database called MonetDB. But before we do that, let's look at the difference between row based (traditional) and column based databases.

Almost all traditional databases store data in the row format, which means in every table, the data is stored one after another in tuple format. In case of column based databases, data is stored column wise (Figure 1).

Column based vs row based databases: The advantages and disadvantages

Let's consider the advantages of column based databases over

row based databases. The former are faster when we need to access all the data of an entire column, because the data is stored together in a single column. Another advantage is that less hard disk access is required and more values can be stored in a block. Column based databases work faster while performing statistical operations like aggregation, summation, etc. This is because data is stored together, so access time and computation time decrease compared to internal storage. When a specific table in the entire column needs to be changed, then a column based database is faster. This is because the entire data in the column is stored in the same block, making it easier to fetch and update it. A column based database is also better in terms of compression, because similar types of values are stored in the same block; in row based databases, the entire tuple, which has different types of values, is stored in one block.

Row based databases are useful when we have more read operations and insertion of records.